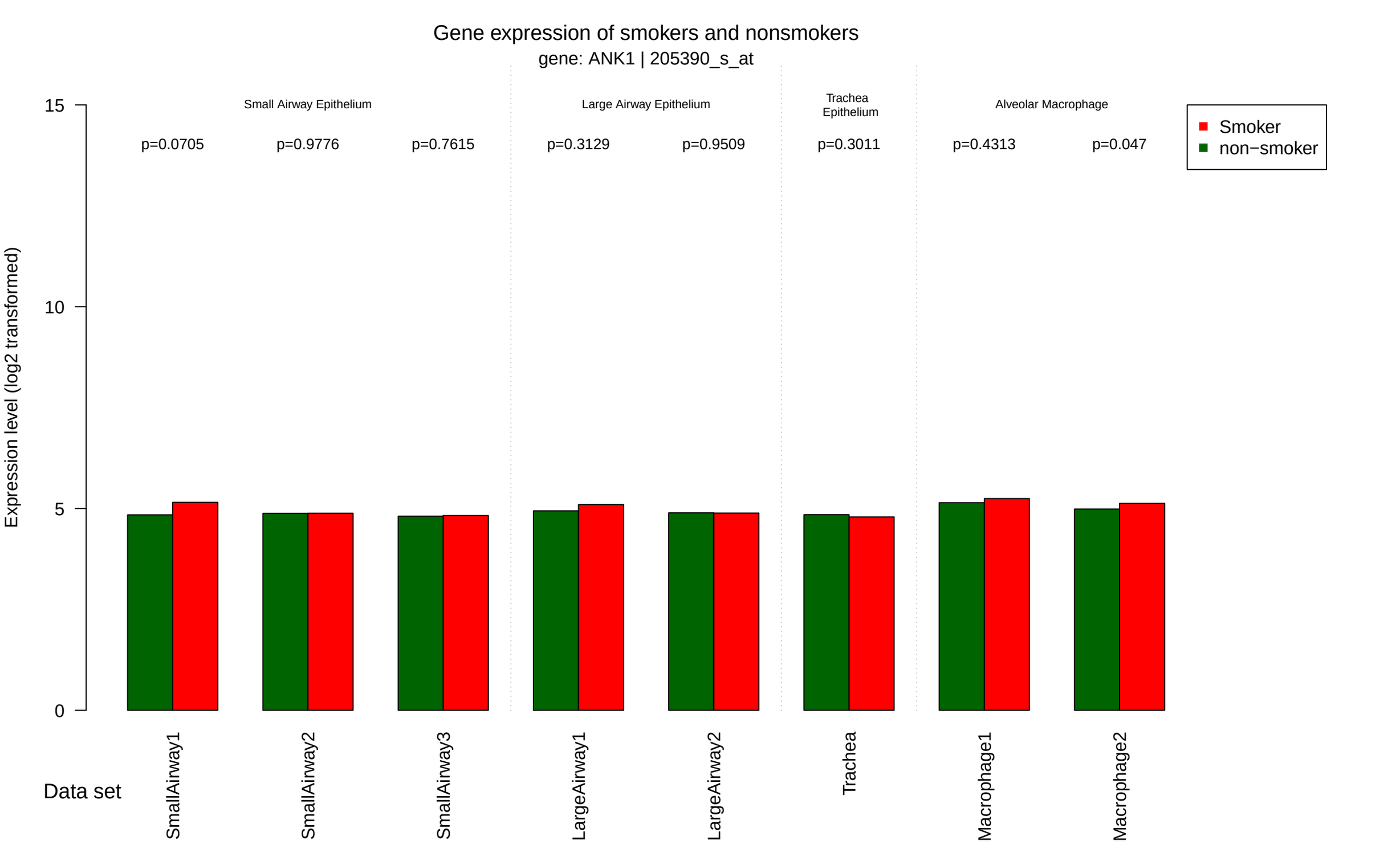
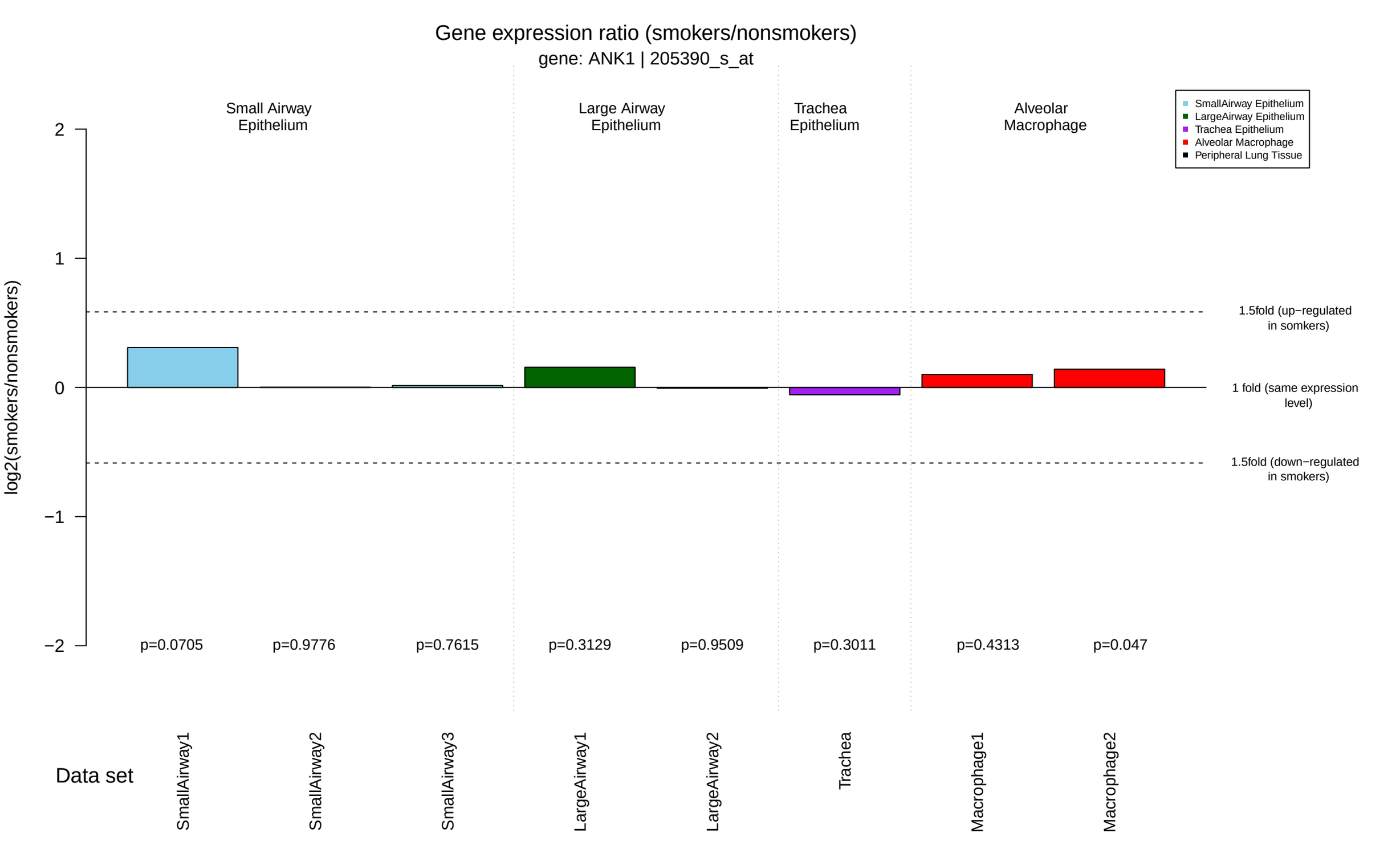
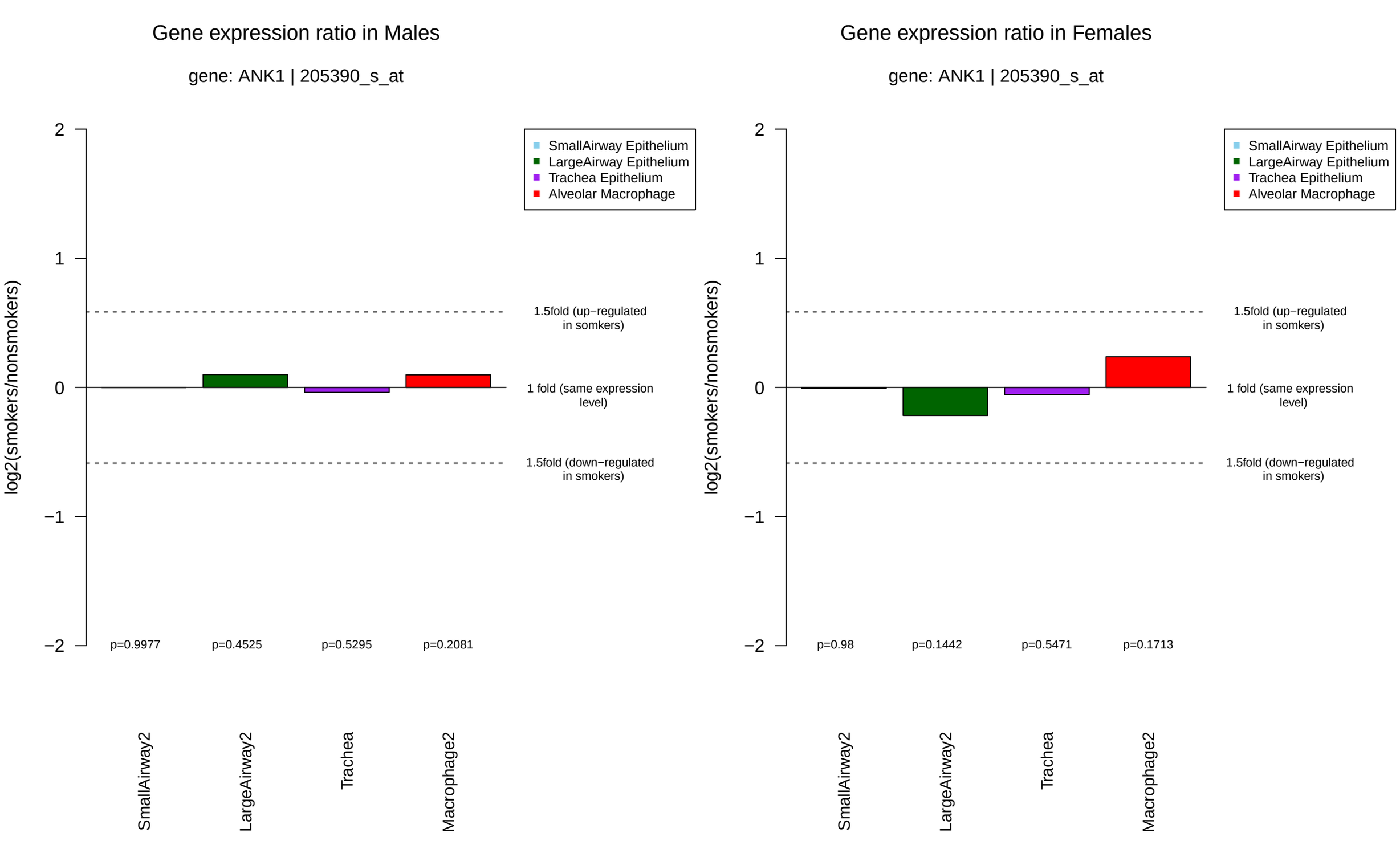
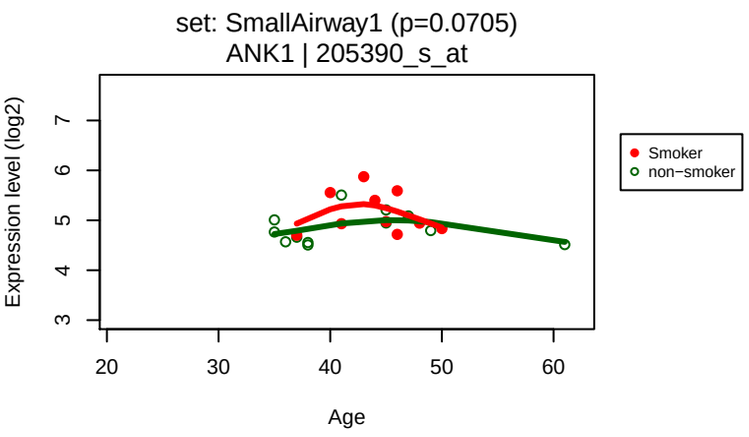
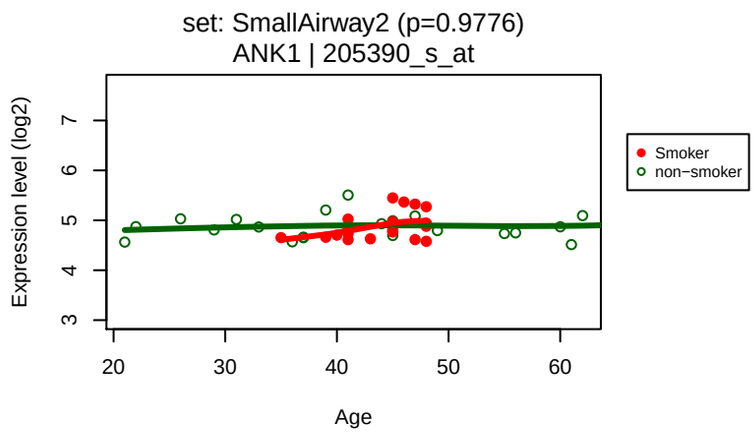
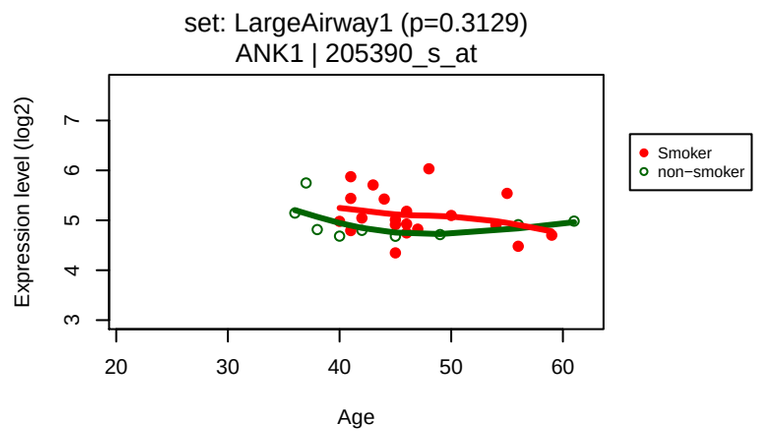

Gene expression of smokers and nonsmokers
gene: ANK1 | 205390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0705
p=0.9776
p=0.7615
p=0.3129
p=0.9509
p=0.3011
p=0.4313
p=0.047
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

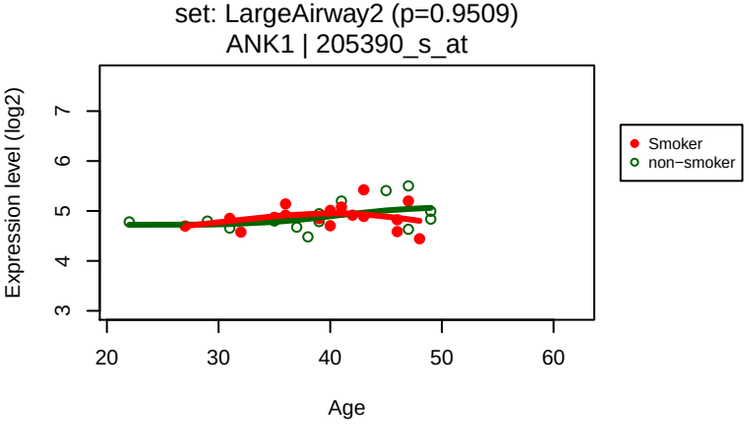

Gene expression ratio (smokers/nonsmokers)
gene: ANK1 | 205390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0705
p=0.9776
p=0.7615
p=0.3129
p=0.9509
p=0.3011
p=0.4313
p=0.047
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

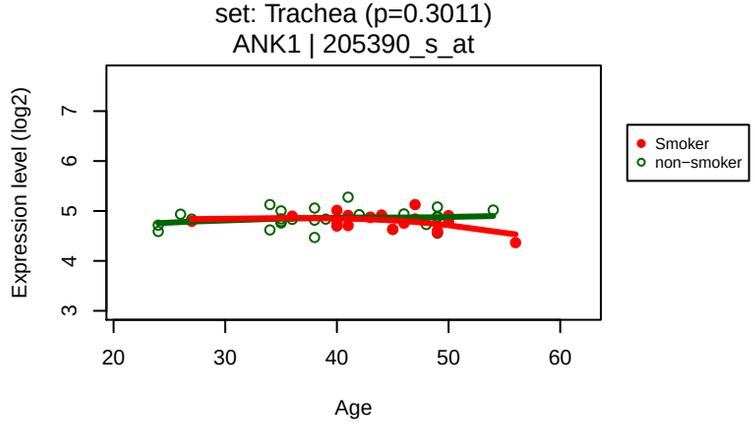

Gene expression ratio in Males
gene: ANK1 | 205390_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9977
p=0.4525
p=0.5295
p=0.2081
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANK1 | 205390_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.98
p=0.1442
p=0.5471
p=0.1713
SmallAirway2
LargeAirway2
Trachea
Macrophage2

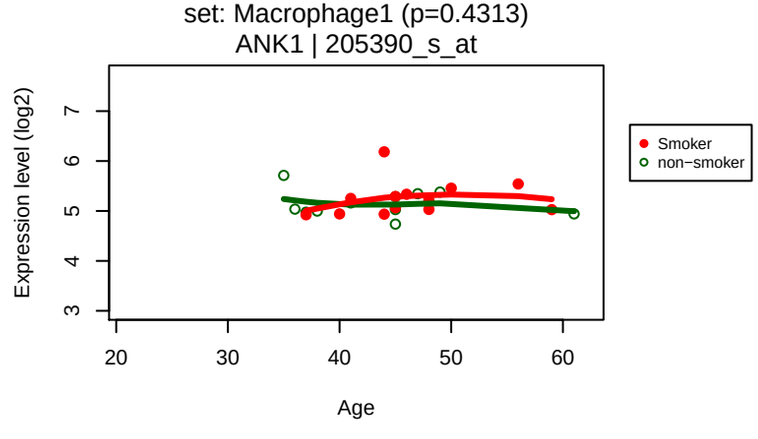

set: SmallAirway1 (p=0.0705)
ANK1 | 205390_s_at
Expression level (log2)
Age
Smoker
non−smoker

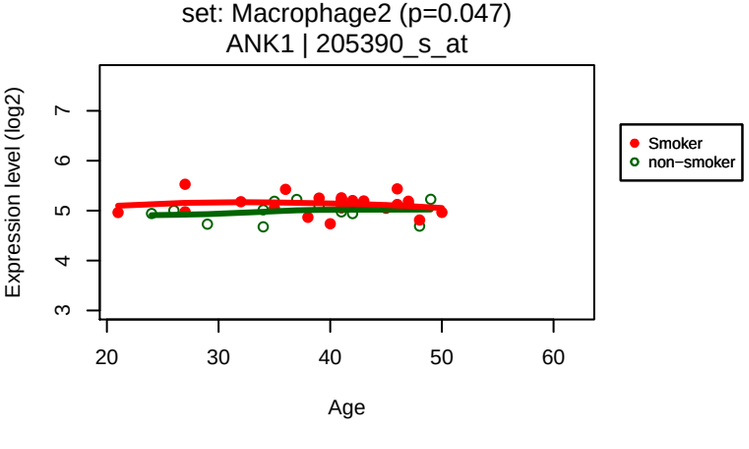

set: SmallAirway2 (p=0.9776)
ANK1 | 205390_s_at
Expression level (log2)
Age
Smoker
non−smoker

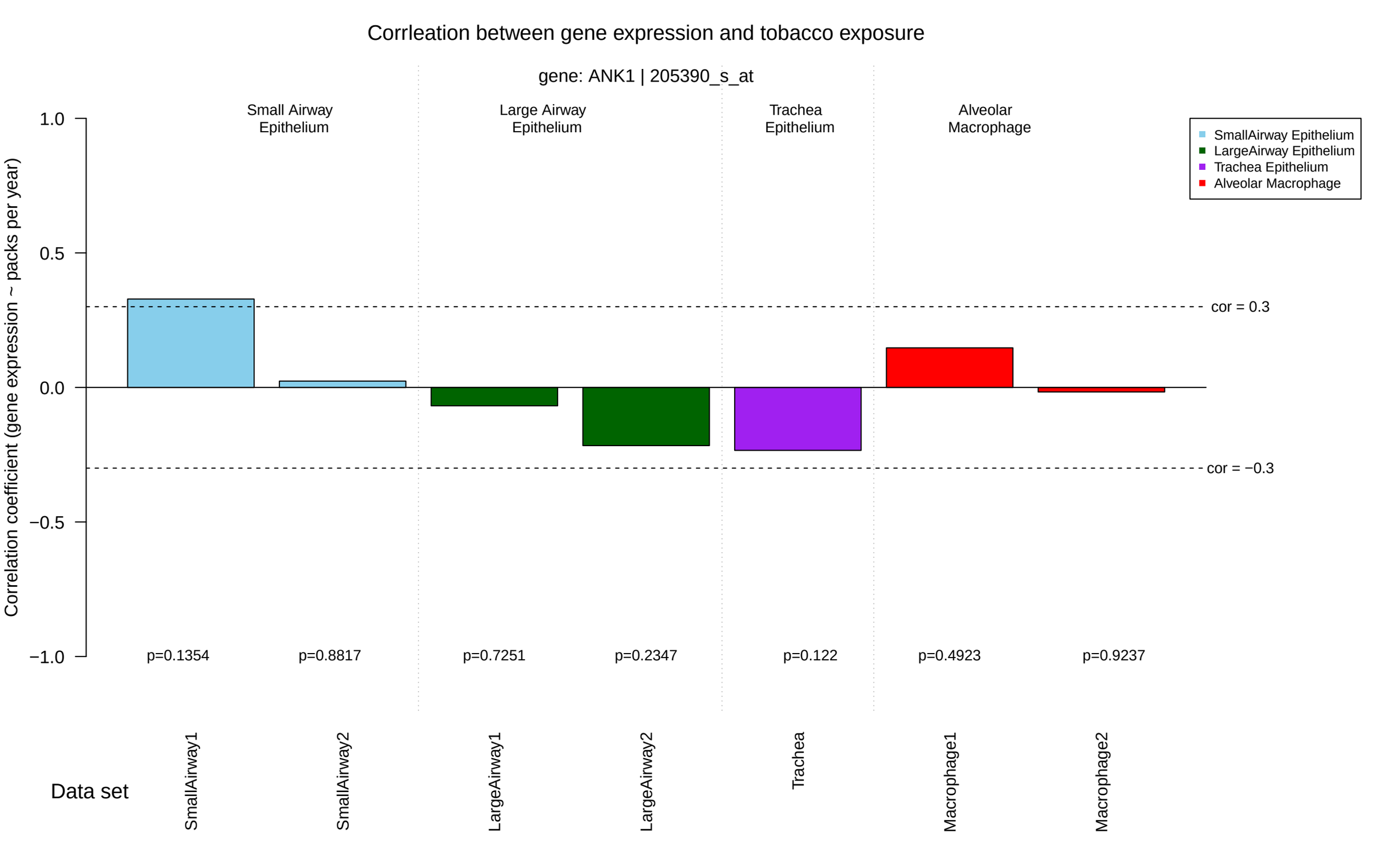

set: LargeAirway1 (p=0.3129)
ANK1 | 205390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9509)
ANK1 | 205390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3011)
ANK1 | 205390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4313)
ANK1 | 205390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.047)
ANK1 | 205390_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANK1 | 205390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1354
p=0.8817
p=0.7251
p=0.2347
p=0.122
p=0.4923
p=0.9237
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2